PROJECT PROPOSAL

AirPulse — Weather & Air Quality Dashboard

Enrollment Numbers: 2503A52049, 2503A52057, 2503A52058
Batch: 02

1. Introduction

AirPulse is a Weather and Air Quality Intelligence Dashboard designed to bring weather conditions and air pollution information together on a single, user-friendly platform. It helps users understand air quality through real-time monitoring, interactive visualizations, location-based analysis, historical trends, and health recommendations.

2. Problem Statement

Weather and air quality information is scattered across multiple platforms and is often difficult for ordinary users to understand. People may struggle to interpret pollution levels, identify high-risk areas, and decide what precautions to take during poor air quality conditions.

AirPulse addresses this problem by combining weather and air quality data into a centralized dashboard with clear visualizations, pollution analysis, and health guidance.

3. Aim and Objectives

Aim:
To develop an interactive dashboard that integrates weather and air quality information to provide useful environmental insights.

Objectives:

- To display real-time AQI and weather conditions.
- To visualize pollutant levels and historical AQI trends.
- To compare air quality across different cities.
- To identify pollution hotspots using interactive maps.
- To provide health recommendations and pollution alerts.
- To analyze the relationship between weather conditions and air pollution.

4. Target Users

- **General Public:** To understand daily weather and pollution conditions.
- **Health-Sensitive Individuals:** To receive guidance during poor air quality conditions.
- **Environmental Authorities:** To monitor pollution patterns and high-risk areas.

5. Proposed Methodology

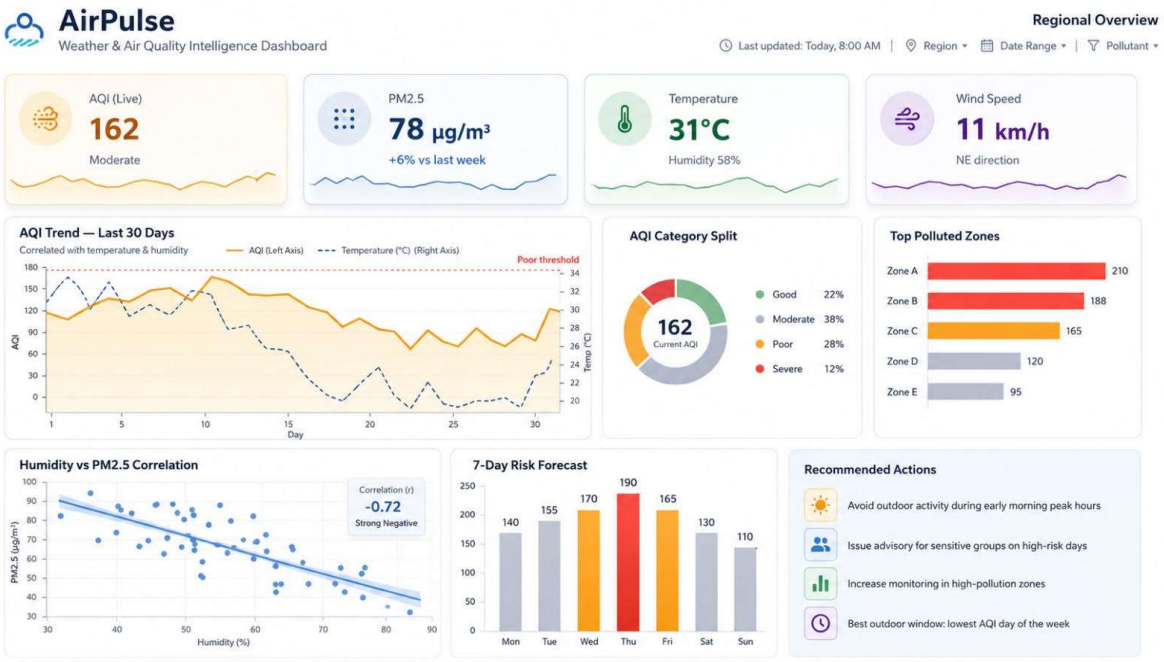
The project will be developed through the following steps:

1. **Data Collection:** Collect weather and air quality data from suitable APIs and reliable sources.
2. **Data Cleaning:** Remove or handle missing, duplicate, and incorrect data to improve data quality.
3. **Data Analysis:** Analyze AQI, pollutant levels, historical trends, and the relationship between weather and air quality.
4. **Data Visualization:** Represent the collected data using charts, graphs, and interactive maps.
5. **Interactive Dashboard:** Develop a user-friendly dashboard to display real-time weather, AQI levels, pollution hotspots, and health advisories.

6. Proposed Features

- AQI monitoring
- Weather details such as temperature, humidity, wind speed, pressure, and rainfall.
- Pollutant monitoring for PM2.5, PM10, NO_2, O_3, CO, and SO_2, where data is available.
- Charts showing historical AQI trends and city-wise comparisons.
- Interactive maps showing pollution hotspots.
- Health advisories and pollution alerts.

7. Expected Dashboard



8. Conclusion

AirPulse aims to make weather and air quality information easier to access and understand. By combining data visualization, interactive maps, and health recommendations, the project will support environmental awareness and informed.